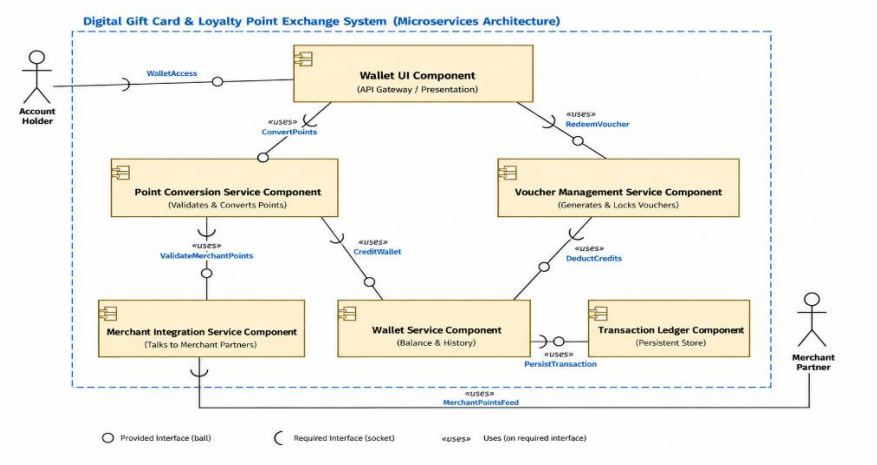
Lab 3: Component Modelling & Architectural Pattern Selection

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Architectural Choice

The system is decomposed into independent, loosely-coupled services Wallet UI, Point Conversion Service, Voucher Management Service, Merchant Integration Service, Wallet Service, and a Transaction Ledger each communicating through well-defined provided/required interfaces (e.g., ConvertPoints, RedeemVoucher, ValidateMerchantPoints). This matches the Microservices style rather than a Layered or Client-Server approach, because the core business capabilities of the system (point validation, conversion, voucher locking, wallet management) are functionally distinct and are triggered by two very different actors Account Holders and Merchant Partners.

Two Reasons for This Choice

- Independent scaling under peak load: FR-001/FR-003 require voucher generation and redemption to be processed within 250 ms (NFR-001) even during high-traffic periods (e.g., festive redemption spikes). Because the Voucher Management Service and Point Conversion Service are separate deployable units, each can be scaled independently based on its own load, without over-provisioning the entire system as a Layered monolith would require.
- Fault isolation across untrusted external integrations: Merchant Partners (FR-004) are external, third-party sources of loyalty-point data of varying reliability. Isolating the Merchant Integration Service as its own component ensures that a slow or failing merchant feed cannot block or crash the Wallet Service or Voucher Management Service a risk that a tightly-coupled Client-Server design would not contain as cleanly.

Security Advantage

Isolating voucher locking logic inside a dedicated Voucher Management Service Component accessed only through its IRedeemVoucher provided interface creates a single, well-guarded choke point for enforcing NFR-002 (single-use voucher locking). No other service can directly mutate voucher state, which shrinks the attack surface for duplicate-redemption or replay attacks compared to a monolithic layer where any business-logic module could touch the voucher table directly.

Performance Benefit

Because the Point Conversion Service and Voucher Management Service run as independent components with their own required/provided interfaces to the Wallet Service (ICreditWallet, IDeductCredits), point conversion and gift-card redemption requests can be processed concurrently on separate service instances rather than contending for a single shared business-logic tier. This directly supports the sub-250 ms latency target in NFR-001 while keeping the ledger consistent through a single Transaction Ledger Component reached via IPersistTransaction.